Industrial Internship Report on

”EXPENSE TRACKER”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). The internship provided an opportunity to gain practical experience and understand the application of technical knowledge. As part of the internship, I worked on a project titled **“Expense Tracker”** using Java. The project is a simple menu-driven application that helps users manage expense records. It provides options to add, display, delete, calculate, and filter expenses. The internship helped me improve my programming, problem-solving, and debugging skills. It was a valuable learning experience that provided practical exposure to software development.

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1 Preface

The Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT) gave me an opportunity to gain practical knowledge and experience in software development.

During the internship, I worked on the **Expense Tracker** project using Java. The project is a simple menu-driven application developed to manage expense records through options such as adding, displaying, deleting, calculating, and filtering expenses.

The internship helped me improve my programming, problem-solving, and debugging skills. I am thankful to the mentors and coordinators for their guidance and support throughout the internship

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

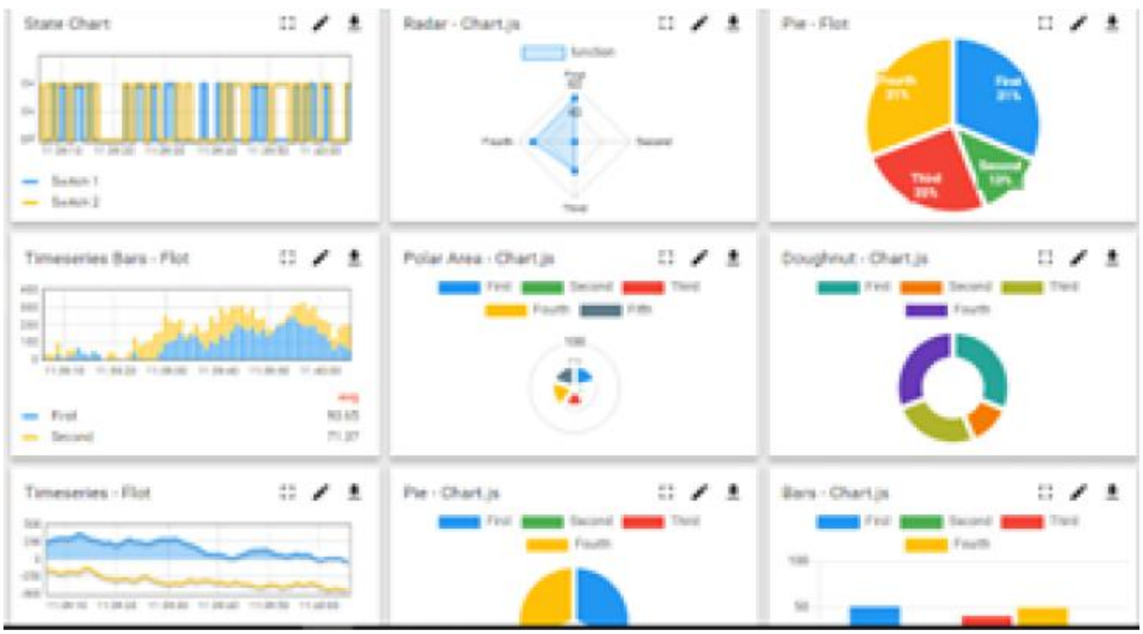
i. UCT IoT Platform ( )

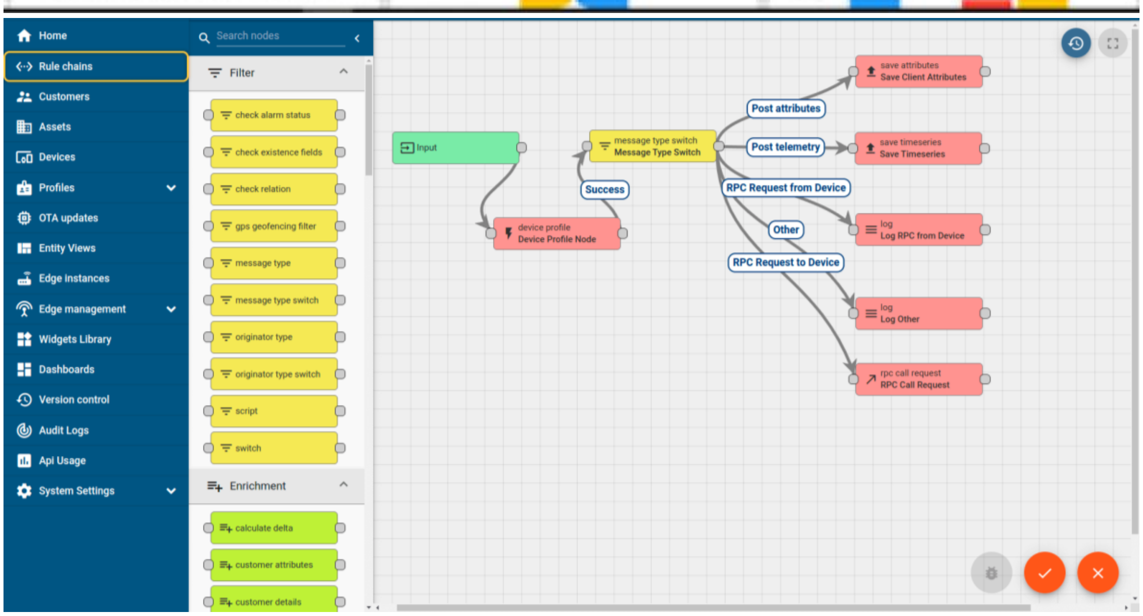
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform ( )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

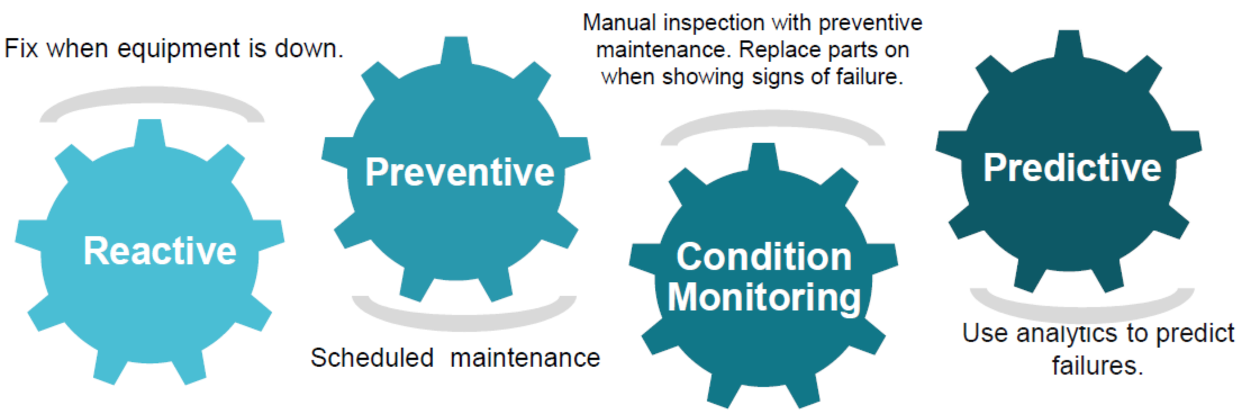
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

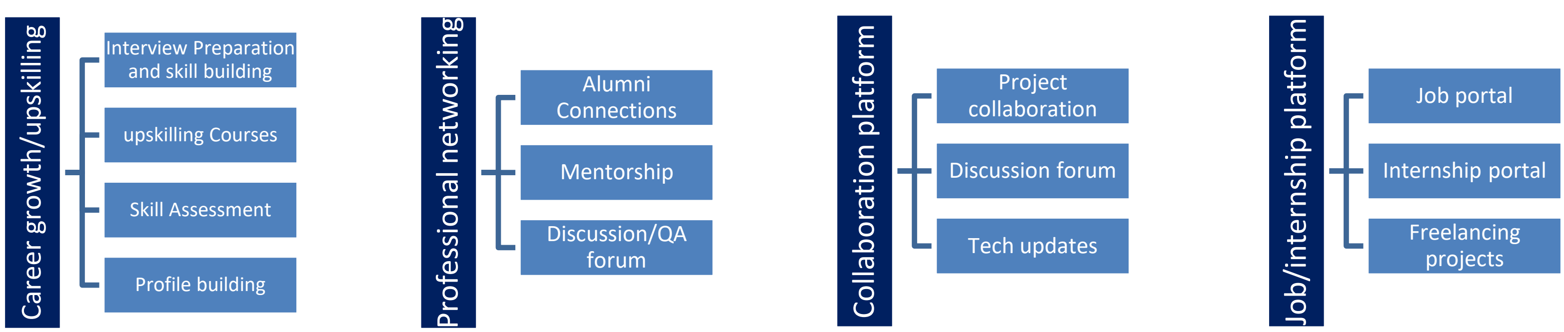
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objectives of the internship and project were to:

- Gain practical experience in software development.
- Improve programming and problem-solving skills.
- Understand the practical application of Java.
- Develop a simple application for managing expense records.
- Gain experience in testing and debugging an application.

2.5 Reference

[1] Java programming documentation and learning resources used for the project.

[2] Java Development Kit (JDK) documentation.

[3] GitHub documentation used for project submission.

2.6 Glossary

Terms	Acronym
Java	Programming language used to develop the project
JDK	Java Development Kit
Scanner	Java class used to take user input
Expense	Money spent or recorded by the user
Category	Group used to classify expenses

3 Problem Statement

Managing daily expenses manually can make it difficult to maintain proper records and calculate total spending. It can also be difficult to find expenses based on their categories.

The **Expense Tracker** provides a simple Java-based solution for managing expense records. It allows users to add, display, delete, calculate, and filter expenses through a menu-driven application.

4 Existing and Proposed solution

Existing Solution

Expenses can be recorded manually using notebooks or spreadsheets. However, manual recording can make it difficult to calculate and manage expenses efficiently.

Proposed Solution

The **Expense Tracker** is a Java-based menu-driven application that provides options to add, display, delete, calculate, and filter expense records.

Value Addition

The application provides a simple and organized way to maintain expense records and calculate total expenses.

4.1 Code submission (Github link)

https://github.com/ramyaragula/upskillcampus/blob/main/ExpenseTracker.java

4.2 Report submission (Github link) :

5 Proposed Design/ Model

The proposed design of the **Expense Tracker** describes how the application works and how the user interacts with its different functions.

5.1 High Level Diagram (if applicable)

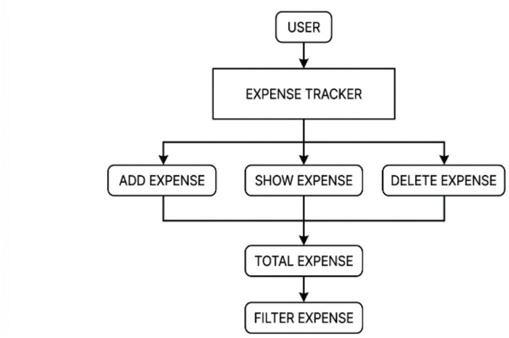
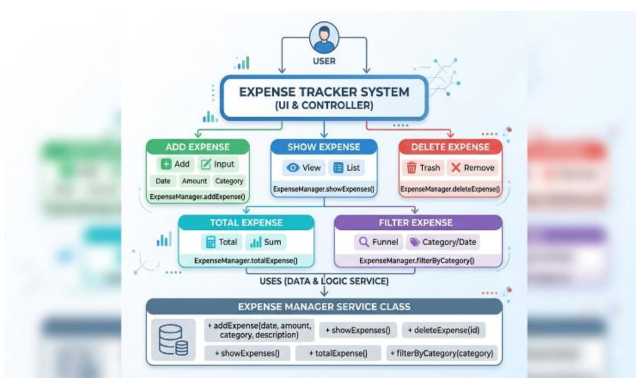


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The Expense Tracker is a console-based Java application. The user interacts with the application through the command-line interface to add, display, delete, calculate, and filter expense records.

6 Performance Test

6.1 Test Plan/ Test Cases

The Expense Tracker was tested to verify that all major operations work correctly.

Test Case	Operation	Expected Result
TC01	Add Expense	Expense is added successfully
TC02	Show Expense	Expenses are displayed
TC03	Delete Expense	Selected expense is deleted
TC04	Total Expense	Total amount is calculated
TC05	Filter Expense	Expenses are filtered by category

6.2 Test Procedure

Each operation was executed using different expense inputs and the obtained results were compared with the expected results.

6.3 Performance Outcome

The tested operations produced the expected results, confirming that the main functions of the Expense Tracker work correctly.

7 My learnings

During the internship, I gained practical knowledge of Java programming and software development. Working on the Expense Tracker project helped me understand how programming concepts can be applied to develop a functional application.

I improved my understanding of classes, objects, methods, user input, conditional statements, and switch-case statements. I also learned how to divide a problem into smaller operations such as adding, displaying, deleting, calculating, and filtering expense records.

The project improved my problem-solving and logical-thinking abilities. While developing and testing the application, I learned how to identify errors, debug the program, and verify whether each operation produced the expected result.

I also gained experience in project documentation and organizing the different stages of a software project. The internship helped me understand the importance of testing, accuracy, and proper implementation while developing an application.

Overall, the internship was a valuable learning experience. It improved my confidence in Java programming and gave me practical exposure to software development. The knowledge and skills gained through this project will help me in developing more advanced applications and in preparing for my future career in the field of computer science

8 Future work scope

The current Expense Tracker is a basic console-based application that provides the essential operations required for managing expenses. However, several improvements can be made in the future to make the application more useful and user-friendly.

A graphical user interface can be added so that users can manage their expenses through an easier and more attractive interface. Database integration can also be implemented to store expense records permanently instead of maintaining them only during program execution.

The application can be further enhanced by adding monthly and yearly expense reports, expense charts, budget management, and spending analysis. Users could also be provided with options to edit existing expenses and generate reports in different formats.

In the future, login and user-account features could be added to provide separate expense records for different users. Cloud storage could also be considered for accessing expense information from different devices.

These improvements can make the Expense Tracker more complete, flexible, and suitable for practical use. The current project provides a basic foundation that can be extended with additional features as future development work.